Unidirectional flow (exact)

Velocity field kinematics:

$$\underline{v} = v_x(y)\,\underline{e}_x \qquad (\Rightarrow \underline{v}\cdot\nabla\underline{v} = \underline{0}\ )$$

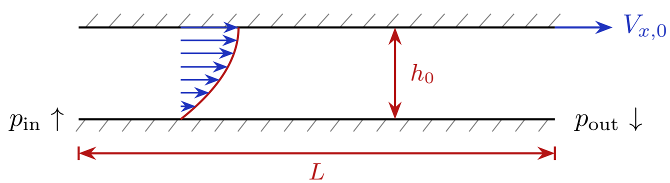
Basic equations:

$$\text{COM:} \qquad 0 = 0$$

$$\text{COLM-}x\text{:} \qquad 0 = -\frac{\partial p}{\partial x} + \mu\frac{\partial^2 v_x}{\partial y^2}$$

$$\text{COLM-}y\text{:} \qquad 0 = -\frac{\partial p}{\partial y}$$

Typical 2D channel geometry ("combined Poiseuille–Couette flow"):



Many other combinations of BCs possible!
Solution strategy:

1. p is not a function of y (from COLM-y)
2. v_x is not a function of x (from kinematics)
3. each term on RHS of COLM-x must be constant:

$$\frac{\partial p}{\partial x} = \frac{dp}{dx} = \text{const.} = \frac{p_{\rm in} - p_{\rm out}}{L} = -\frac{\Delta p}{L}$$

$$v_x(y) = \frac{h_0^2}{2\mu}\left(-\frac{dp}{dx}\right)\left[\frac{y}{h_0} - \left(\frac{y}{h_0}\right)^2\right] + V_{x,0}\,\frac{y}{h_0}$$

4. $Q = \int_0^{h_0} v_x(y)\,dy$ leads to flow rate–pressure *drop* relation (**done!**)

Lubrication theory (approx)

Velocity field kinematics:

$$\underline{v} = v_x(x,y)\,\underline{e}_x + v_y(x,y)\,\underline{e}_y \qquad (\text{ with } \underline{v}\cdot\nabla\underline{v} \approx \underline{0}\ )$$

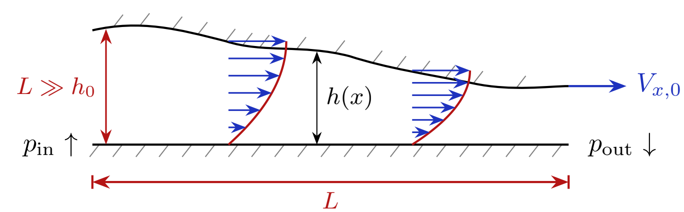
Basic equations:

$$\text{COM:} \qquad \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0$$

$$\text{COLM-}x\text{:} \qquad 0 \approx -\frac{\partial p}{\partial x} + \mu\frac{\partial^2 v_x}{\partial y^2}$$

$$\text{COLM-}y\text{:} \qquad 0 \approx -\frac{\partial p}{\partial y}$$

Typical 2D channel geometry:



Solution strategy:

1. p is not a function of y (COLM-y)
2. v_x *is* a function of x; solve COLM-y for v_x in terms of $\partial p/\partial x$:

$$v_x(x,y) = \frac{h(x)^2}{2\mu}\left(-\frac{\partial p}{\partial x}\right)\left[\frac{y}{h(x)} - \left(\frac{y}{h(x)}\right)^2\right] + V_{x,0}\,\frac{y}{h(x)}$$

3. $Q = \int_0^{h(x)} v_x(x,y)\,dy$ leads to flow rate–pressure *gradient* (**unknown!**)
4. write COM as $\partial v_y/\partial y = -\partial v_x/\partial x$
   (a) integrate over *all* y and use BCs to get ODE for $p(x)$
   (b) integrate up to *arbitrary* y to find $v_y(x,y)$ (optional)